

Assistance Needs Observed in Human–Human Coassembly

851 annotated assistance events



DETAILED EVENT CATEGORIES

- **PHYSICAL SUPPORT**
 - Part Out of Reach — 172/851 (20.2%)
 - Workpiece Stabilization — 36/851 (4.2%)
- **PART SEARCH & TASK DECISIONS**
 - Unable to Locate Part — 124/851 (14.6%)
 - Uncertain About Next Step — 123/851 (14.5%)
- **ASSEMBLY ERROR CORRECTION**
 - Incorrect Part Selection — 96/851 (11.3%)
 - Assembly-Order Violation — 63/851 (7.4%)
 - Incorrect Part Orientation — 63/851 (7.4%)
 - Incorrect Screwdriver / Bit — 57/851 (6.7%)
 - Missed Fastener — 22/851 (2.6%)
- **CONVERSATIONAL SUPPORT**
 - Confirmation / Affirmation — 52/851 (6.1%)
 - Other Conversational Support — 43/851 (5.1%)